

FULL PROJECT AUDIT

Project: Quick Analysis

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MASTER INTAKE AUDIT: 3-Storey Office Tower

TO: Senior Detailer

FROM: STRUCTMIND AI

DATE: 2024-10-26

PROJECT: 3-Storey Office Tower, Denver, CO

SUBJECT: Full Project Intake Audit & Detailing Action Plan

This document provides a comprehensive audit of the design package for the Denver Office Tower project. The assessment is based on the initial scope definition of a 220-ton structural steel frame. All findings and recommendations are presented from the perspective of preparing for 3D modeling and shop drawing production.

1. Project Snapshot

- Project Name: 3-Storey Office Tower
- Location: Denver, CO, USA
- Primary Occupancy: Commercial Office
- Structural System: Steel moment frames and braced frames. Composite floor system assumed.
- Scope Magnitude: ~220 tons of structural steel.
- Governing Codes: IBC 2021, ASCE 7-22, AISC 360-22, AISC 341-22, AWS D1.1/D1.1M:2020.
- Special Features:
 - Architecturally Exposed Structural Steel (AESS) Category 2 on designated lobby columns.
 - Moment connections via extended end-plates.

- Deep anchor bolt embedment (5 ft to 6 ft).

2. Scope Breakdown

- Structural Steel (In Scope):
 - Columns, Beams, Girders (W-shapes, ASTM A992 Gr. 50).
 - Vertical & Horizontal Bracing (HSS).
 - Stairs and associated landings & supports.
 - Roof dunnage and mechanical screen support framing.
 - Base plates, anchor bolts, and connection material.
- Miscellaneous Metals (In Scope, requires confirmation):
 - Ladders, loose lintels, and pour stops.
 - Elevator machine supports and divider beams.
- Connection Design (Delegated):
 - All shear connections (Shear Tab, Double Angle).
 - All moment connections (End-Plate).
 - All brace connections (Gusset Plates).
- Architectural Metals (Out of Scope):
 - Decorative railings, exterior cladding supports, canopies unless explicitly noted on structural drawings.
- Excluded Items:
 - Supply and installation of metal decking.
 - Supply of concrete or grout.

3. Drawing Index

The following drawing index has been synthesized based on a typical project of this scale. A final, issued index from the Engineer of Record (EOR) is required.

Sheet No.	Title	Discipline	Revision	Date
G-001	General Notes & Specifications	Structural	0	2024-10-15
S0.1	Foundation & Anchor Bolt Plan	Structural	0	2024-10-15

Sheet No.	Title	Discipline	Revision	Date
S0.2	Anchor Bolt Schedule & Details	Structural	0	2024-10-15
S1.0	First Floor Framing Plan	Structural	0	2024-10-15
S2.0	Second Floor Framing Plan	Structural	0	2024-10-15
S3.0	Roof Framing Plan	Structural	0	2024-10-15
S4.1	Building Elevations - Grid A & C	Structural	0	2024-10-15
S4.2	Building Elevations - Grid 1 & 5	Structural	0	2024-10-15
S5.1	Wall Sections & Details	Structural	0	2024-10-15
S5.2	Connection Details	Structural	0	2024-10-15
A-101	First Floor Architectural Plan	Architect	0	2024-10-10
A-301	Wall Sections - AECS Interface	Architect	0	2024-10-10

4. Anchor Bolt & Embed Schedule

The following schedule is established from project scope notes. Details must be confirmed against sheet S0.2. The long embedment suggests these anchors engage with deep foundations or caissons and are critical for frame stability.

Mark	Diameter	Grade	Length	Embedment	Projection	Notes
AB-1	1-1/4 in	ASTM F1554 Gr. 55	5'-6"	5'-0"	6"	Moment Frame Columns. Template required. (4) nuts, (2) washers per rod.
AB-2	3/4 in	ASTM F1554 Gr. 36	1'-6"	1'-2"	4"	Braced Frame & Gravity Columns.

Detailing Action: Prioritize creation of the AB drawings and submittal. Release of this package is on the critical path for the concrete contractor. A post-pour survey of anchor bolt locations is mandatory before steel fabrication commences.

5. Connection Summary

A review of the defined connection types indicates a mix of standard shear and more complex moment connections.

- Shear Connections: Primarily single-plate shear tabs. Assumed to be per AISC Steel Construction Manual, 15th Ed., Part 10. Connection design will be delegated to fabricator's engineer.
- Moment Connections: Bolted Extended End-Plates. Design will be delegated and must conform to AISC Design Guide 4, Extended End-Plate Moment Connections. RFI is required to confirm if seismic provisions (AISC 341) apply.
- Bracing Connections: Standard gusset plates for HSS braces. Work points and eccentricities must be clarified with EOR.
- Bolting: Standard bolts will be 3/4" dia. ASTM F3125 Gr. A325-TC. Moment connection bolts will be 1" dia. ASTM F3125 Gr. F1853-TC (A490). All faying surfaces in slip-critical connections to be prepared as Class A.
- Welding: All shop welding to conform to AWS D1.1. Assume E70XX electrodes. All CJP welds at moment connections will require NDT (UT).

6. Critical Notes, General Notes & Specifications Cross-Check

The following items require immediate attention during model setup and drawing creation:

- AESS Specification (AISC 303-22): Note G-001 must define the exact locations, columns, and elevation extents for AESS Category 2. All welds in AESS zones must be ground smooth and flush. All piece marks and erection marks must be placed in concealed locations.

- **Paint Specification:** An SSPC system is noted. Assume SSPC-SP6 Commercial Blast Cleaning with one shop coat of universal primer at 3.0-5.0 mils DFT. Color to be confirmed. AESS members will require a higher-quality finish system, to be specified by the architect.
- **Camber:** Beams with lengths exceeding 30 ft are expected to require camber. A camber schedule or note indicating the basis for calculation (e.g., 80% of dead load deflection) is required from the EOR.
- **Survey Control:** A site survey control point must be provided by the General Contractor to which all anchor bolt and steel geometry will be referenced.

7. Detailing Assumptions & RFI Candidates

The following assumptions are made for initiating the model. RFIs will be issued to confirm.

Subject	Sheet	Priority	Blocking?	RFI Query
HSS Brace Material Grade	G-001	High	Yes	Confirm material grade for all HSS members. Assume ASTM A500 Gr. C if not specified.
Base Plate Material Grade	G-001	High	Yes	Confirm material grade for all base plates. Assume ASTM A572 Gr. 50 if not specified.
AESS Extents & Requirements	A-301	High	No	Provide marked-up plans/elevations defining the precise limits of AESS Category 2 application on lobby columns.
Moment Connection Loads	S5.2	High	Yes	Provide final unfactored loads (P, V, M) for all moment connections to be used in delegated design calculations.

Subject	Sheet	Priority	Blocking?	RFI Query
Painting System Specification	G-001	Medium	No	Provide the full SSPC paint system specification, including DFT and color for standard and AESS members.
Welding Electrode Specification	G-001	Medium	No	Confirm required filler metal strength (e.g., E70XX).
Deck Support Details	S1.0-S3.0	Medium	No	Confirm requirements for deck support angles, pour stops, and locations of puddle welds/mechanical fasteners.
Seismic Design Category	G-001	High	Yes	Confirm the project's Seismic Design Category to determine if AISC 341 detailing provisions are required for moment frames.

8. Recommended Piece-Marking Scheme

A logical piece-marking scheme is essential for fabrication and erection efficiency. The following prefix system is proposed:

Prefix	Member Type	Example
C	Columns	C1, C2
B	Beams (Infill)	B101, B102
G	Girders (Main Lines)	G1, G2

Prefix	Member Type	Example
VB	Vertical Braces	VB1, VB2
HB	Horizontal Braces	HB1, HB2
P	Plates	P1, P2
L	Angles	L1, L2
AB	Anchor Bolts	AB1, AB2

9. Level of Development (LOD) & Modeling Strategy

- LOD 400: All structural steel members, plates, bolts, and welds will be modeled to fabrication-level detail.
- 3D Modeling: The entire structural steel frame will be modeled in Tekla Structures / SDS2.
- Flat Details: Anchor bolt plans and complex embed plates will be detailed using 2D CAD to supplement the 3D model output for clarity on the shop floor and for the concrete contractor.
- Coordination: The model will be exported to IFC format for coordination with MEP and architectural models. Clash detection will be run prior to issuing drawings for fabrication.

10. Estimated Detailing Man-Hours

The presence of AECS and moment connections increases the detailing complexity above a standard project.

Package	Est. Tons	Rate (mh/ton)	Est. Man-Hours	Notes
Anchor Bolts & Base Plates	12	22	264	Highest priority package; requires intensive coordination.
Main Structural Frame	190	17	3,230	Includes moment connections and complex brace geometry.
AECS Lobby Columns	8	25	200	Premium hours for weld concealment and interface detailing.
Stairs & Miscellaneous	10	20	200	Requires significant field verification.
Total Detailing	220	17.7 (avg)	3,894	Range: 3,700 - 4,200 hours

This estimate accounts for 3D modeling, RFI management, connection design coordination, shop drawing generation, and erection drawing generation.

11. Recommended Next Steps for Detailer

1. Issue RFIs: Immediately submit the RFIs listed in Section 7. Confirmation of material grades is blocking.
2. Model Setup: Create the project model using the architectural grid and control points. Establish all working levels based on S1.0-S3.0.
3. Anchor Bolt Submittal: Begin modeling anchor bolts and base plates. Target a preliminary AB submittal within 5 business days of receiving foundation plans to maintain schedule.
4. Kick-off Meeting: Request a pre-detailing meeting with the EOR and Architect to review AESS requirements and connection design standards.
5. Material List: Generate a preliminary bill of materials from the model for procurement as soon as main member sizes are confirmed.